

Domáci úkol 3.2

Wednesday 25 February 2026 17:54

$$\langle f | g \rangle = \int_1^5 f(x) g(x) dx$$

a) $V = \text{Span} \left\{ \underset{v_1}{1}, \underset{v_2}{x}, \underset{v_3}{x^2} \right\} \rightarrow \text{ON báze } V.$

$$u_1 = \frac{1}{\|1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{2}$$

OG projekce
 $w_2 = \left\langle \frac{1}{2}, x \right\rangle \frac{1}{2} = \frac{1}{2} \int_1^5 x dx \cdot \frac{1}{2} = \frac{12}{4} = 3$

$$u_2 = \frac{v_2 - w_2}{\|v_2 - w_2\|} = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} (x-3)$$

$$w_3 = \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}}{4} (x-3), x^2 \right\rangle \frac{\sqrt{3}}{4} (x-3) =$$

$$= \frac{1}{4} \int_1^5 x^2 dx + \frac{3}{16} \int_1^5 x^3 - 3x^2 dx (x-3) = \frac{31}{3} \cdot \frac{1}{4} + \frac{3}{16} \cdot \frac{2}{32} (x-3) = 6x - \frac{23}{3}$$

$$u_3 = \frac{v_3 - w_3}{\|v_3 - w_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{256}{45}}} = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right)$$

ON báze V je $B = (0.5, 0.43x - 1.3, 0.42x^2 - 2.52x + 3.21)$

b) OG projekce $\sin x$ na $\text{Span} \left\{ \frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \right\}$

OG projekce
 $w = \left\langle \frac{1}{2}, \sin x \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}}{4} (x-3), \sin x \right\rangle \frac{\sqrt{3}}{4} (x-3) + \left\langle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}), \sin x \right\rangle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$

$$w = \frac{1}{4} \int_1^5 \sin x dx + \frac{3}{16} \int_1^5 (x-3) \sin x dx (x-3) + \frac{45}{256} \int_1^5 (x^2 - 6x + \frac{23}{3}) \sin x dx (x^2 - 6x + \frac{23}{3}) \approx$$

$$\approx -0.0525(x^2 - 6x + 7.6) - 0.647(x-3) + 0.064 = -0.0525x^2 - 0.332x + 1.6025$$

c) $\| \sin x - (ax^2 + bx + c) \| = \sqrt{\int_1^5 (\sin x - (-0.0525x^2 - 0.332x + 1.6025))^2 dx} \approx 0.31879$